CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

# Chapter 23: Graphical Representation (Histograms and Ogives)

Every formula and result of the chapter on one sheet

| | | |
|---|---|---|
| 1 | Class mark | $\frac{\text{lower limit} + \text{upper limit}}{2}$ |
| 2 | Class size | Upper limit $-$ lower limit (continuous classes) |
| 3 | Adjustment factor | $\frac{\text{lower limit of a class} - \text{upper limit of the previous class}}{2}$ |
| 4 | Histogram | Rectangles on continuous classes; height $=$ frequency; no gaps |
| 5 | Mode from a histogram | Foot of the perpendicular from the crossing point of the two diagonal lines in the tallest rectangle |
| 6 | Ogive | Plot (upper class limit, cumulative frequency); join with a smooth curve |
| 7 | Starting point of an ogive | (lower limit of the first class, 0) |
| 8 | Median from an ogive | Value at cumulative frequency $\frac{N}{2}$ |
| 9 | Lower quartile $Q_1$ | Value at cumulative frequency $\frac{N}{4}$ |
| 10 | Upper quartile $Q_3$ | Value at cumulative frequency $\frac{3N}{4}$ |
| 11 | Inter-quartile range | $Q_3 - Q_1$ |
| 12 | Number above a value | $N-$ cumulative frequency at that value |